Central Bank Communication as Alternative Data
Topic Modeling Study of BIS Speeches

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Abstract

This project studies whether central bank speeches can be transformed into interpretable textual topics using unsupervised Natural Language Processing methods. I analyze 18,922 speeches from the Bank for International Settlements central bank speeches corpus over 1997–2025 and compare three topic modeling approaches: TF-IDF with Non-negative Matrix Factorization, Latent Dirichlet Allocation on count vectors, and BERTopic with sentence-transformer embeddings. The main methodological conclusion is that unsupervised topic models are useful for exploratory macro-financial text analysis, but their temporal interpretation requires caution because topic prevalence can reflect both genuine changes in communication and changes in the institutional composition of the corpus. As an exploratory extension, I construct a dictionary-based hawkish-dovish tone score to measure changes in restrictive or accommodative monetary-policy vocabulary.

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1 Introduction

Central bank communication is a major source of information for financial markets. Speeches by central bankers often contain signals about inflation risks, the monetary policy stance, financial stability concerns, growth prospects, and macroeconomic uncertainty. Unlike policy rates or inflation data, these speeches are unstructured textual data. They are therefore a natural candidate for Natural Language Processing (NLP) methods in a macro-financial context, where investors increasingly use alternative data to monitor changes in policy priorities.

My choice of project is motivated by the connection between NLP, quantitative finance, and macroeconomic analysis. As I am working in hedge funds on alpha research, I chose this topic because I found it interesting for my career. The objective is not to apply topic models mechanically, but to study whether central bank speeches can be transformed into interpretable textual indicators. The research question is: **Can unsupervised NLP methods extract interpretable topics from central bank speeches, how do these topics evolve over time, and can a simple lexical score capture changes in hawkish-dovish communication?**

The contribution is fourfold. First, I construct a cleaned modeling corpus from BIS central bank speeches and document its statistical limitations. Second, I compare three topic modeling approaches with different assumptions: NMF on TF-IDF vectors, LDA on count vectors, and BERTopic on sentence embeddings. Third, I evaluate whether the recovered topics are economically interpretable, temporally plausible, and robust across modeling choices. Fourth, I build a small exploratory dictionary-based hawkish-dovish tone score to complement the topic models with a monetary-policy tone indicator, as an extension of my initial topic modeling project.

2 Related Work

Central bank communication has become an important object of empirical research because monetary authorities influence expectations not only through policy decisions, but also through speeches, minutes, press conferences and forward guidance. [1] show that the language used by the Federal Open Market Committee contains information about both current economic conditions and future policy guidance. In a related contribution, [2] use computational linguistics to study deliberation within the FOMC and show that central bank texts can be analyzed as structured empirical data rather than only as qualitative documents. This project follows the same general idea: central bank speeches are treated as alternative textual data that may reveal changes in macro-financial priorities over time. However, unlike papers that focus on causal effects or market reactions, my objective is exploratory and descriptive: I study whether unsupervised NLP methods can recover interpretable topics from a large corpus of BIS speeches.

From an NLP perspective, topic modeling provides a family of unsupervised methods for summarizing large document collections. Latent Dirichlet Allocation (LDA), introduced by [3], represents each document as a mixture of latent topics and each topic as a distribution over words. This probabilistic interpretation is useful for macro-financial documents because one speech can discuss several issues, such as inflation, financial stability and growth. However, LDA relies on a bag-of-words representation and can produce diffuse topics when documents share a large common vocabulary. Non-negative Matrix Factorization (NMF), introduced by [4], provides an alternative matrix-factorization view. When applied to TF-IDF vectors, NMF often produces sparse and interpretable topics because terms that are frequent in a document but less common in the corpus receive higher weight. This motivates the use of both LDA and NMF in the project: LDA is a standard probabilistic benchmark, while NMF is expected to produce sharper exploratory topics.

More recent topic modeling approaches use dense semantic representations rather than sparse word-count matrices. Sentence embeddings, such as those introduced by [5], map sentences or documents into continuous vector spaces where semantic similarity can be measured more directly than with exact word overlap. BERTopic [6] combines transformer-based embeddings, dimensionality reduction, clustering and class-based TF-IDF to produce topic representations. Including BERTopic is therefore useful because central bank speeches may contain semantically similar passages expressed with different vocabularies. At the same time, this approach has limitations for the present corpus: BIS speeches are long documents, and document-level embeddings may capture institutional or geographical style in addition to economic content.

Finally, this project is related to the literature on dictionary-based tone measurement in finance and monetary policy. [7] show that generic sentiment dictionaries can be inappropriate for financial texts

because words may have domain-specific meanings. In the context of central bank communication, [8] construct a field-specific measure of ECB communication and distinguish between dovish and hawkish policy content. This motivates the exploratory hawkish-dovish extension of the project. The dictionary score used here is deliberately simple and transparent: it is not intended to replace a supervised stance classifier, but to complement the topic models by measuring whether inflation-tightening vocabulary becomes more frequent in recent speeches.

The literature therefore suggests two complementary questions. First, can unsupervised topic models recover economically meaningful themes from central bank communication? Second, do these themes evolve in a way that is consistent with major macro-financial periods? The present project addresses these questions descriptively on the BIS speeches corpus.

3 Data and Preprocessing

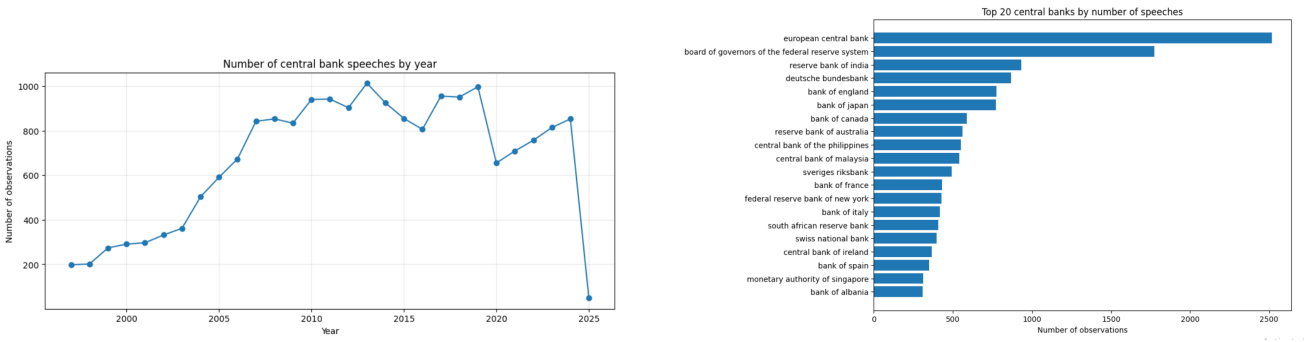
3.1 Corpus description and exploratory analysis

The project uses the BIS central bank speeches dataset available on Hugging Face under the name `samchain/bis_central_bank_speeches`. The raw dataset contains 19,376 observations and five variables: `a speech description, the full speech text, the associated central bank, the year and the month`. The corpus covers 1997–2025 and contains speeches from 118 central banks or monetary institutions.

The dataset is appropriate for topic modeling because it is large, temporally rich and composed of long documents. The median raw speech length is 2,389 words. However, the corpus is also imbalanced: some years contain many more speeches than others, and the top 10 central banks account for about 51% of observations, while the top 20 account for about 71%.

Table 1: Main exploratory statistics.

Quantity	Value
Raw observations	19,376
Non-missing speech texts	19,211
Missing speech texts	165
Distinct central banks	118
Period covered	1997–2025
Mean words per speech	2,718
Median words per speech	2,389
Exact duplicate non-missing texts	40



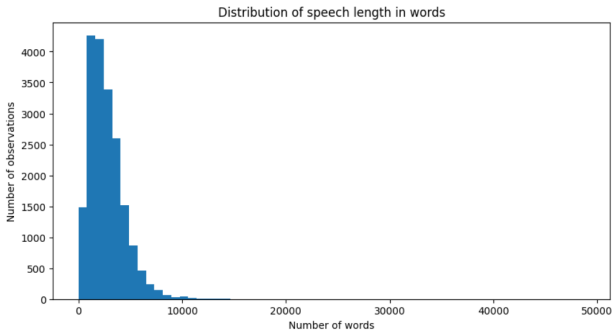
(a) Number of speeches by year.

(b) Top 20 central banks.

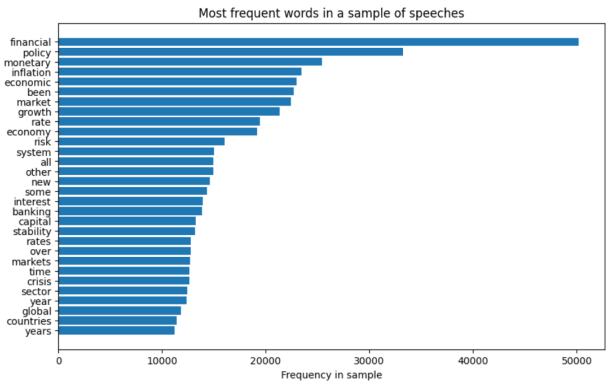
Figure 1: Temporal and institutional coverage of the BIS speeches corpus.

Figure 1 highlights two important properties of the corpus. First, the number of speeches varies strongly over time, so the temporal analysis must use normalized topic proportions rather than raw topic counts. The sharp decline in 2025 should be interpreted cautiously, as the year is likely incomplete at the time of data collection. Second, the corpus is institutionally concentrated: the top institutions contribute a large fraction of the speeches. In fact, European central banks account for a large share of the speeches in the corpus as shown in Figure 1b. This is important for topic modeling because unsupervised models may capture the vocabulary and communication style of highly represented institutions, rather than only purely economic themes.

Speech length is also heterogeneous, as shown in Figure 2. Very short documents are removed before modeling because they contain too little context for reliable topic assignment. The word-frequency plot confirms the domain relevance of the corpus, with frequent terms such as *financial*, *policy*, *monetary*, *inflation*, *market*, *growth*, *rate*, *risk* and *stability*. Generic frequent terms motivate the use of a domain-specific stopword list.



(a) Speech length distribution.



(b) Frequent words in a sample.

Figure 2: Document length and lexical structure of the corpus.

To interpret the temporal evolution of topics, I group years into broad macro-financial periods, as reported in Table 2. This aggregation makes the yearly topic dynamics easier to interpret economically and reduces noise in year-by-year fluctuations. For example, if a banking-regulation topic becomes more prevalent during 2007–2009, this is consistent with the global financial crisis; if an inflation topic rises after 2021, this is consistent with the recent inflation-tightening period. These periods are not exact structural breaks for every country in the corpus, but they provide a useful economic framework to summarize changes in topic prevalence.

Table 2: Macro-financial periods used for temporal interpretation.

Period	Years	Interpretation
Pre-GFC	1997–2006	Pre-crisis macro-financial environment
Global Financial Crisis	2007–2009	Financial crisis and liquidity stress
Euro crisis / post-GFC	2010–2014	Regulation and euro-area sovereign stress
Low rates / QE	2015–2019	Low rates and unconventional policy
Covid	2020–2021	Pandemic shock and emergency policy response
Inflation tightening	2022–2025	Inflation shock and monetary tightening

3.2 Preprocessing

The preprocessing step constructs the final modeling corpus. I remove missing and empty texts, documents with severe encoding artifacts, exact duplicates, and speeches with fewer than 200 raw words. The encoding-quality filter removes 227 speeches. For classical bag-of-words models, I create `clean_text`.

The pipeline lowercases the text, removes URLs and email addresses, normalizes whitespace, protects economically meaningful expressions such as *monetary policy*, *financial stability*, *interest rate*, *exchange rate*, *euro area*, *Federal Reserve*, *Basel III* and *Covid-19*, then applies **spaCy tokenization, lemmatization and part-of-speech filtering**. Only nouns, proper nouns, adjectives and verbs are retained.

I use standard English stopwords and domain-specific stopwords such as *central*, *bank*, *banks*, *speech*, *governor*, *year* and *time* (this is detailed in notebook 02). These terms are structurally frequent and would dominate topics without adding discriminative information.

For BERTopic, I keep a less aggressively cleaned representation, `bertopic_text`, because sentence embeddings benefit from more natural linguistic context.

Table 3: Preprocessing summary.

Step	Observations
Raw dataset	19,376
After removing missing or empty texts	19,185
After encoding-quality filter	18,958
After removing duplicates and short raw texts	18,929
Final modeling corpus	18,922

The final corpus represents 97.66% of the original dataset, still covers 1997–2025 and includes 118 central banks. The median cleaned length is 1,093 words and the mean cleaned length is 1,255 words.

4 Models and Strategy

4.1 TF-IDF and NMF

For NMF, each speech is represented with a TF-IDF vector. TF-IDF assigns high weights to terms that are frequent in a document but relatively specific in the corpus. For a term t and a document d, the weight is:

$$\mathrm{tfidf}(t,d) = \mathrm{tf}(t,d) \times \log\left(\frac{N}{\mathrm{df}(t)}\right),$$

where $\mathrm{tf}(t,d)$ is the frequency of term t in document d, N is the number of documents, and $\mathrm{df}(t)$ is the number of documents containing the term. This representation is well suited to exploratory document analysis because it downweights terms that are frequent everywhere in the corpus.

NMF then factorizes the non-negative document-term matrix X into two non-negative matrices:

$$X \approx WH, \qquad \min_{W,H\geq 0} \|X - WH\|_F^2.$$

Here, W is the document-topic matrix and H is the topic-term matrix. The non-negativity constraint makes the decomposition interpretable: each document is represented as an additive mixture of topics, and each topic is represented as an additive mixture of terms.

4.2 Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) is a probabilistic topic model. Unlike NMF, which is applied to TF-IDF weights, LDA is fitted on a count-based document-term matrix because it models word occurrences. Each document d is associated with a topic distribution θ_d, and each topic k is associated with a word distribution ϕ_k. For each word position in a document, a topic is sampled from θ_d, and then a word is sampled from the corresponding topic-word distribution ϕ_k. LDA therefore represents each document as a probability distribution over topics and each topic as a probability distribution over words.

4.3 BERTopic

BERTopic is an embedding-based topic modeling approach. Instead of directly modeling a sparse document-term matrix, BERTopic maps documents into dense semantic vectors using a sentence-transformer model. In this project, I use `all-MiniLM-L6-v2`, which produces 384-dimensional embeddings. The pipeline embeds speeches, reduces the embedding space with UMAP, clusters the reduced representations with HDBSCAN, and represents each cluster using class-based TF-IDF. HDBSCAN can also assign ambiguous documents to an outlier topic, denoted by -1.

BERTopic is included as a modern semantic extension, but it has an important limitation in this project. BIS speeches are long documents, often longer than the maximum effective input length of transformer-based embedding models. Therefore, full-document embeddings may not represent all parts of each speech equally and may over-represent the beginning of speeches, where speaker, institution and location information often appear. This limitation is important for interpreting the institutional clusters found by BERTopic.

4.4 Evaluation strategy

Because topic modeling is unsupervised, there is no ground-truth label against which topics can be evaluated with standard supervised metrics such as accuracy or F1-score. I therefore evaluate topic quality using a combination of quantitative diagnostics and qualitative validation.

First, I use model-specific fit diagnostics. For NMF, reconstruction error measures how well WH approximates the TF-IDF matrix. For LDA, perplexity and log-likelihood measure how well the model fits the observed word counts. These metrics are useful for comparing candidate models, but they tend to improve mechanically as model complexity increases and should not be used alone.

Second, I evaluate topic quality through coherence and diversity. Coherence measures whether the top words of a topic tend to appear together in the corpus, while topic diversity measures whether different topics rely on distinct top words rather than repeating the same vocabulary. Third, I perform qualitative validation by inspecting the top words and the most representative speeches for each topic. A topic is considered useful only if its dominant words and representative speeches support a coherent economic interpretation. Finally, I evaluate temporal plausibility by checking whether topic prevalence evolves consistently with major macro-financial periods.

5 Empirical Protocol and Results

5.1 NMF: protocol, model selection and results

For NMF, I build a TF-IDF matrix from the `clean_text` representation. I use unigrams and bigrams because many central banking concepts are multi-word expressions, such as *monetary policy*, *financial stability*, *exchange rate* and *euro area*. The vectorizer keeps terms appearing in at least 20 documents, removes terms appearing in more than 85% of documents, and caps the vocabulary at 10,000 features.

The number of topics is not known in advance. I therefore estimate NMF models with:

$$K \in \{4, 6, 8, 10, 12, 15, 20\}.$$

This grid compares coarse decompositions, medium-size decompositions and more fragmented specifications.

Table 4: NMF evaluation by number of topics.

K	Recon. error	Decr./topic	Diversity
4	126.14	–	0.950
6	124.83	0.655	0.883
8	123.75	0.540	0.900
10	122.72	0.511	0.870
12	121.86	0.431	0.842
15	120.63	0.411	0.847
20	118.84	0.357	0.810

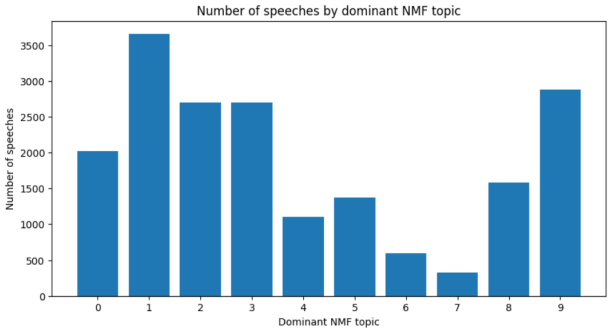
Table 5: Final NMF topics for $K = 10$.

Topic	Manual label	Dominant terms
T0	Inflation and monetary policy	inflation, monetary policy, rate, price, target, expectations
T1	Banking regulation and stability	risk, banking, capital, crisis, Basel, regulation, supervision
T2	Euro area and monetary union	euro, euro area, ECB, European, union, fiscal
T3	Global growth and exchange rates	country, exchange, global, growth, trade, currency
T4	Payments and digital finance	payment, digital, technology, fintech, innovation, CBDC
T5	Federal Reserve and US policy	Federal Reserve, FOMC, Fed, rate, labor, federal funds
T6	Climate risk and sustainable finance	climate, climate change, green, transition, carbon, emissions
T7	Islamic finance and Malaysia	Islamic finance, Malaysia, shariah, sukuk, takaful
T8	Japan and monetary easing	Japan, prices, growth, rate, firms, demand
T9	Financial inclusion and development	financial inclusion, BSP, Philippine, microfinance, education

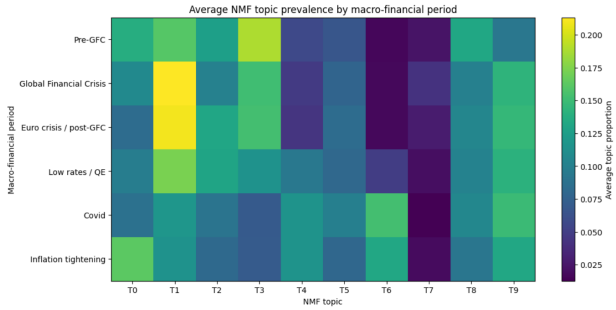
The reconstruction error decreases monotonically with K, but the marginal improvement per added topic declines (as shown in Table 4). Topic diversity also tends to decrease for larger specifications, indicating more overlap between topics. I select $K = 10$ because it provides the best compromise between reconstruction quality, diversity and interpretability. This choice is not based only on the numerical metrics: models with $K = 4$ or $K = 6$ merge several macro-financial themes, while models with $K = 15$ or $K = 20$ increasingly fragment the corpus into institution-specific or country-specific topics. The $K = 10$ solution separates the main economic themes while keeping the topic set readable.The words obtained for all candidate values of K are reported in Appendix A. They support this choice: small values of K merge distinct macro-financial themes, while larger values produce more fragmented and institution-specific topics.

The topics are economically meaningful (Table 5). Some correspond to broad macro-financial themes, such as inflation, banking regulation, payments, climate finance and global exchange-rate issues. Others capture institutional or regional structure, such as the Federal Reserve, the euro area, Japan, Islamic finance and financial inclusion. This is expected in an international corpus with strong institutional

imbalance.



(a) Dominant NMF topic.



(b) Topics by macro period (Temporal evolution).

Figure 3: NMF topic distribution and macro-financial interpretation.

Figure 3 summarizes the dominant-topic distribution and average topic prevalence by macro-financial period. Banking regulation and financial stability is the most frequent dominant topic, followed by financial inclusion, euro-area communication and global macro-financial themes.The macro-period heatmap is particularly useful because it connects the topics to the temporal dimension of the project. The macro-period heatmap shows that banking regulation becomes more prominent around the global financial crisis and the euro crisis/post-GFC period. Payments and digital finance become more visible during the low-rate/QE period, while climate-related communication becomes much more visible in the Covid and post-Covid period. These patterns suggest that the NMF topics are not only interpretable in terms of top words, but also temporally plausible.

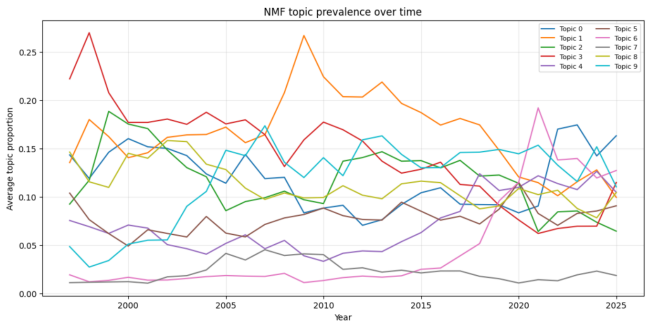


Figure 4: NMF topic prevalence over time. Each line reports the yearly average topic proportion across speeches.

Figure 4 provides a yearly view. The global growth and exchange-rate topic is important in the early part of the sample. The banking-regulation topic rises around the global financial crisis and remains high during the post-crisis years. Payments and digital finance increase after the 2010s. Climate is almost absent in the early years but rises sharply in the recent period. Inflation and monetary policy also rise again during the inflation-tightening period. These patterns are economically plausible, but they should be interpreted as changes in communication emphasis rather than as direct measures of monetary policy stance.

5.2 LDA: protocol, model selection and results

For LDA, I use the same `clean_text` representation but replace TF-IDF with a `CountVectorizer`, because LDA models word counts. The vectorizer uses unigrams and bigrams, keeps terms appearing in at least 20 documents, removes terms appearing in more than 85% of documents, and limits the vocabulary to 10,000 features. The resulting matrix has 18,922 rows and 10,000 columns, with a sparsity of 93.81%.

I estimate LDA models with the same topic grid as NMF:

$$K \in \{4, 6, 8, 10, 12, 15, 20\}.$$

I keep the default symmetric Dirichlet priors for the document-topic distribution θ_d and the topic-word distribution ϕ_k, corresponding to the LDA hyperparameters α and η in scikit-learn.

K	Perp.	Log-like.	Mean UMass coh.
4	2297.68	-201.21M	-0.315
6	2166.43	-199.68M	-0.387
8	2089.48	-198.74M	-0.355
10	2029.84	-197.99M	-0.475
12	1998.48	-197.58M	-0.510
15	1930.44	-196.68M	-0.567
20	1881.61	-196.01M	-0.518

Table 6: LDA evaluation by number of topics.

Topic	Top words	Label / interpretation
T0	growth, productivity, labour, investment, GDP, employment, wage	Growth and labour markets
T1	risk, banking, capital, regulation, supervision, Basel, stability	Banking regulation and financial stability
T2	reserve, federal reserve, liquidity, asset, bond, balance sheet	Federal Reserve liquidity and asset purchases
T3	credit, loan, debt, household, mortgage, lending, housing	Credit, debt and housing markets
T4	euro area, European, ECB, union, crisis, fiscal, member	Euro area and monetary union
T5	price, rate, growth, inflation, demand, Japan, outlook	Inflation, activity and Japan outlook
T6	inflation, monetary policy, target, expectation, price, stability, forecast	Inflation targeting and monetary policy framework
T7	payment, digital, banking, technology, transaction, Islamic, innovation	Payments, digital finance and Islamic finance
T8	exchange, foreign, international, exchange rate, capital, currency, trade	Global exchange rates and international macro
T9	development, climate, finance, public, government, support, community	Development, climate and public finance

Table 7: Final LDA topics for $K = 10$.

For LDA, lower perplexity indicates a better fit to the observed word counts, while higher log-likelihood is preferred. Since log-likelihood values are negative, "higher" means less negative. In Table 6, perplexity decreases and log-likelihood increases as K grows, which is expected because larger models have more flexibility. However, these metrics are computed in-sample and tend to favor more complex models. They are therefore used only as relative diagnostics, not as predictive validation metrics.

UMass coherence evaluates whether the most probable words of a topic tend to co-occur in the same documents. Higher values, meaning less negative values here, are preferred. However, UMass coherence does not improve monotonically with K. In fact, small models have better coherence scores but merge several distinct economic themes, while larger models improve statistical fit but produce more diffuse or fragmented topics. I therefore select $K = 10$ because it provides a readable compromise between statistical fit, coherence, topic granularity and qualitative interpretability. The words obtained for all candidate values of K are reported in Appendix B.

Compared with NMF, LDA mixes related vocabularies more often. For example, payments and Islamic finance appear in the same topic, and climate is grouped with development and public finance. This reflects the probabilistic nature of LDA and its reliance on raw word co-occurrences.

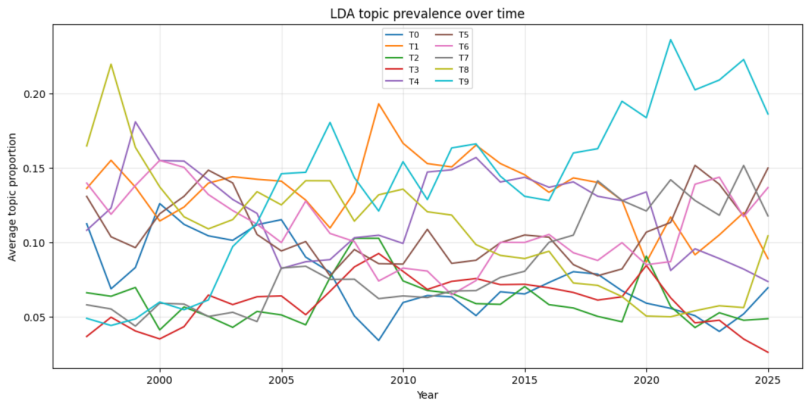


Figure 5: LDA topic prevalence over time. Each line reports the yearly average topic probability across speeches.

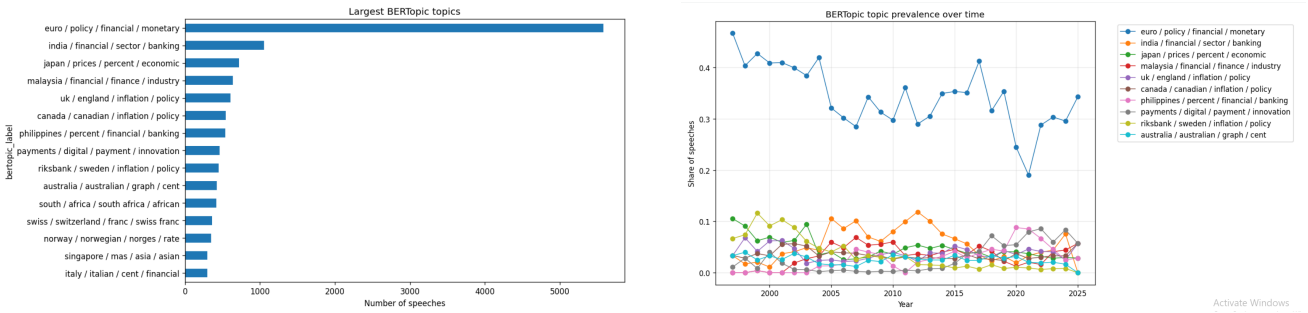
Figure 5 should be interpreted as changes in broad lexical emphasis rather than causal evidence of policy shifts. The global exchange-rate topic is relatively important early in the sample, banking regu-

lation becomes more visible around the crisis period, and the development/climate/public-finance topic becomes dominant in recent years. This broadly confirms the NMF evidence while illustrating that LDA topics are less sharply separated.

5.3 BERTopic: protocol, model selection and results

For BERTopic, the number of topics is not fixed in advance but is determined by HDBSCAN. I use the less aggressively cleaned `bertopic_text` representation because sentence-transformer embeddings benefit from natural linguistic context. Each speech is embedded with `all-MiniLM-L6-v2`, producing 384-dimensional vectors. UMAP is then used with `n_neighbors=15`, `n_components=5`, `min_dist=0.0` and cosine distance. HDBSCAN is configured with `min_cluster_size=100`, approximately 0.5% of the corpus, and `min_samples=10`. These parameters avoid very small unstable clusters while allowing ambiguous speeches to remain outliers.

The final BERTopic model identifies 38 non-outlier clusters and assigns 10.75% of speeches to the outlier topic. The model obtains a C_v coherence score of 0.446 and a top-10 topic diversity score of 0.468, indicating moderate coherence and moderate lexical overlap. This is expected because central bank speeches share a common monetary and financial vocabulary.



(a) Largest BERTopic clusters. **(b)** BERTopic topic prevalence over time.

Figure 6: BERTopic largest-topic distribution and temporal evolution.

Figure 6(a) shows that BERTopic is dominated by one very large cluster, *euro - policy - financial - monetary*, containing 5,582 speeches. This should not be interpreted as a narrow topic. It is a broad monetary-financial communication cluster, reflecting the common semantic core of many BIS speeches. The other large clusters often combine a geographical or institutional marker with an economic vocabulary, such as *uk - england - inflation - policy*, *canada - canadian - inflation - policy*, *malaysia - financial - finance - industry*, or *philippines - financial - banking*. This is an important result: BERTopic captures not only themes, but also country-specific and institution-specific communication styles.

Figure 6(b) shows that the largest cluster accounts for roughly 20% to 45% of non-outlier speeches depending on the year. These shares are much higher than NMF or LDA topic proportions because BERTopic performs hard clustering: each speech is assigned to one main cluster, while NMF and LDA represent speeches as mixtures of topics. The decline of the largest cluster around 2020–2021 suggests that the corpus becomes less concentrated around one broad monetary-financial discourse and more differentiated across institutions and themes. BERTopic is therefore useful as a semantic robustness extension: it shows that the BIS corpus is structured by macro-financial themes, but also by geography, institutional identity and communication style.

6 Comparison of Topic Models

Table 8: Comparison of the three topic modeling approaches.

Model	Representation	Main contribution	Main limitation
NMF	TF-IDF matrix	Sharp macro-financial topics	Sensitive to preprocessing
LDA	Count matrix	Probabilistic benchmark	More diffuse topics
BERTopic	Sentence embeddings	Semantic and institutional clusters	Hard clustering; long-document embeddings

The three models provide complementary views of the corpus. NMF gives the clearest macro-financial topics and the most readable temporal narratives. It identifies themes such as inflation, banking regulation, euro-area communication, payments, climate finance and financial inclusion. Its temporal evolution is economically plausible: banking regulation becomes more salient around the global financial crisis, payments and digital finance increase after the 2010s, climate communication rises in the recent period, and inflation-related communication increases again during the inflation-tightening period (heatmap).

LDA broadly confirms several NMF themes but produces more diffuse topics. Because it models raw word co-occurrences, related vocabularies are sometimes grouped together: for example, payments and Islamic finance appear in the same topic, and climate is mixed with development and public finance. LDA is therefore useful as a probabilistic robustness check, but its temporal curves should be interpreted as broad lexical emphasis rather than sharply separated economic topics.

BERTopic provides a different interpretation. It does not mainly recover compact macro-financial topics; instead, it reveals semantic, geographical and institutional structure. This difference is not a contradiction. NMF factorizes TF-IDF weights, LDA models word counts, and BERTopic clusters dense document embeddings. The models therefore do not estimate the exact same object. Taken together, they show that central bank communication is structured both by economic narratives and by the identity of the institutions producing the speeches.

7 Hawkish-Dovish Tone Extension

To complement the topic models, I construct an exploratory dictionary-based hawkish-dovish tone score. Topic models identify what central banks talk about; the tone score measures whether the vocabulary is relatively more *hawkish* or *dovish*. In monetary-policy language, hawkish communication refers to restrictive language associated with inflation concerns, price stability and monetary tightening. Dovish communication refers to accommodative language associated with policy support, liquidity provision, monetary easing and weak economic activity. This score is not a supervised stance classifier, but a transparent lexical proxy for monetary-policy communication.

The score uses two components. First, I count single-token terms in the lemmatized `clean_text`. Hawkish terms include words related to inflation pressure and policy restriction, such as *inflation*, *price*, *tightening*, *restrictive*, *wage*, *target* and *credibility*. Dovish terms include words related to accommodation and weak activity, such as *easing*, *support*, *liquidity*, *cut*, *unemployment*, *recession*, *recovery* and *downside*. Second, I match multi-word expressions in `bertopic_text`, such as *monetary tightening*, *inflation pressures*, *quantitative easing*, *asset purchases* and *downside risks*.

For each speech d, I compute:

$$\text{Tone}_d = 1000 \times \frac{H_d - D_d}{N_d},$$

where H_d and D_d are the weighted hawkish and dovish counts, and N_d is the number of cleaned tokens. Positive values indicate more hawkish than dovish vocabulary, while negative values indicate more dovish than hawkish vocabulary. Since the raw score is right-skewed, I winsorize it at the 1st and 99th percentiles before aggregation. Importantly, the score should be interpreted in relative terms: a positive average does not mean that all speeches are restrictive, but that hawkish terms are more frequent than dovish terms according to this specific dictionary. Therefore, the comparison across periods is more informative than the absolute level of the score.

Period	Speeches	Mean tone	Median tone
Pre-GFC	3,708	24.83	16.20
Global Financial Crisis	2,517	16.74	9.61
Euro crisis / post-GFC	4,716	11.43	6.74
Low rates / QE	4,521	11.97	6.93
Covid	1,264	6.00	3.07
Inflation tightening	2,196	27.51	14.85

Table 9: Average hawkish-dovish tone by macro-financial period.

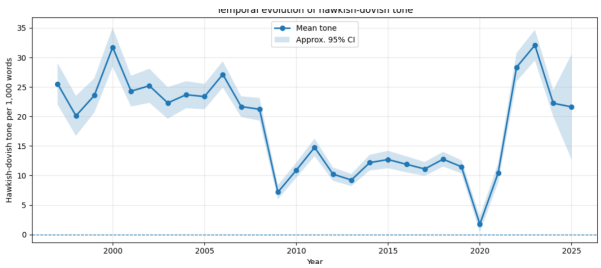


Figure 7: Temporal evolution of the hawkish-dovish tone score. The score is a winsorized net hawkish lexical count per 1,000 words.

Table 9 shows that all period averages are positive. This reflects the fact that central bank speeches

frequently use terms related to inflation, price stability and policy targets. The result should therefore not be read as evidence that all periods are absolutely hawkish. The relevant information is the relative variation across periods. Covid has the lowest average score, consistent with more accommodative language, liquidity provision and policy support during the pandemic. The inflation-tightening period has the highest score, consistent with the renewed importance of inflation, price stability and restrictive monetary-policy language after 2021–2022. The Pre-GFC period is also relatively hawkish, likely reflecting the importance of inflation-targeting and price-stability vocabulary in central bank communication before the global financial crisis.

Figure 7 confirms the period-level pattern. The tone score rises sharply in 2022 and 2023, during the inflation-tightening period, and then declines slightly in 2024. This suggests that the increase in inflation-related topics identified by NMF and LDA is accompanied by a more hawkish lexical tone. As validation, the unigram-only version of the score is highly correlated with the main score, with Pearson correlation 0.989 and Spearman correlation 0.984. A face-validity comparison also shows that the inflation-tightening period is much more hawkish than the Covid period: the mean score rises from 6.00 to 27.51, with a statistically significant difference under both Welch and Mann–Whitney tests.

This extension remains deliberately simple. A full sentiment or monetary-policy stance analysis would be a separate project. With more time, I would construct a validated hawkish-dovish dictionary specific to central bank communication, manually label a subset of speeches, and compare this dictionary score with a supervised classifier or a transformer-based stance model. The present score should therefore be interpreted as an exploratory lexical indicator, useful for complementing the topic models but not sufficient to measure policy stance on its own.

8 Conclusion, Limitations and Extensions

This project shows that central bank speeches can be converted into interpretable textual indicators using unsupervised NLP methods. The main result is that the corpus is structured along two dimensions. First, it contains clear macro-financial themes, such as inflation, banking regulation, euro-area communication, payments, climate finance and financial inclusion. These are best captured by NMF, which provides the sharpest and most readable topics. Second, the corpus is also shaped by institutional and geographical patterns, which are especially visible in BERTopic. LDA confirms several of the main themes, but with more diffuse probabilistic topics. The hawkish-dovish score provides an additional validation: the recent inflation-tightening period is associated with a more hawkish lexical tone than the Covid period.

Overall, the analysis supports the research question: unsupervised NLP methods can extract meaningful topics from central bank communication, and these topics evolve in economically plausible ways over time. However, the results should be interpreted as changes in observed communication emphasis, not as causal measures of monetary-policy stance.

Limitations and possible extensions.

- **Institutional imbalance.** The corpus is dominated by a limited number of central banks. Future work should compute topic dynamics within homogeneous groups of institutions or reweight speeches by central bank.
- **Unsupervised evaluation.** Topic labels remain partly subjective. A stronger validation would combine coherence, representative speeches and expert/manual annotations on a small sample.
- **Long-document embeddings.** BERTopic is applied at the speech level, although speeches are long. A better approach would split speeches into chunks, embed each chunk, and aggregate chunk-level topics at the document level.
- **Tone score.** The hawkish-dovish score is a transparent lexical proxy, not a supervised stance model. A stronger extension would use a validated central-bank dictionary or manually labeled speeches to train and evaluate a classifier.

A NMF Candidate Topics

Table 10 reports the highest-weighted words obtained by NMF for each candidate number of topics. This appendix supports the choice of $K = 10$: small values of K merge several macro-financial themes, whereas larger values of K increasingly split the corpus into more institution-specific or country-specific components.

Table 10: Top NMF terms for all candidate values of K. Each topic is represented by its seven highest-weighted terms.

K	Topics and top terms	
4	T0	inflation, rate, price, monetary, monetary policy, growth, percent, ...
	T1	risk, capital, crisis, banking, institution, credit, liquidity, ...
	T2	euro, euro area, area, european, ecb, monetary, union, ...
	T3	payment, service, digital, technology, sector, inclusion, bsp, ...
6	T0	inflation, monetary, monetary policy, rate, price, target, inflation target, ...
	T1	risk, crisis, banking, capital, institution, basel, supervision, ...
	T2	euro, euro area, area, european, ecb, union, monetary, ...
	T3	growth, rate, cent, price, japan, increase, investment, ...
	T4	payment, service, digital, technology, inclusion, financial inclusion, islamic, ...
	T5	federal, federal reserve, reserve, percent, fomc, fed, labor, ...
8	T0	inflation, monetary, monetary policy, rate, price, target, inflation target, ...
	T1	risk, banking, crisis, capital, institution, basel, regulation, ...
	T2	euro, euro area, area, european, ecb, union, monetary, ...
	T3	growth, rate, cent, price, japan, increase, investment, ...
	T4	payment, digital, service, technology, inclusion, financial inclusion, bsp, ...
	T5	federal, federal reserve, reserve, percent, fomc, fed, labor, ...
	T6	climate, climate change, green, risk, transition, climate relate, carbon, ...
	T7	islamic, islamic finance, islamic financial, malaysia, shariah, finance, sukuk, ...
10	T0	inflation, monetary, monetary policy, rate, price, target, inflation target, ...
	T1	risk, banking, capital, crisis, basel, institution, regulation, ...
	T2	euro, euro area, area, european, ecb, union, monetary, ...
	T3	cent, country, exchange, global, growth, rate, exchange rate, ...
	T4	payment, digital, service, technology, fintech, innovation, card, ...
	T5	federal, federal reserve, reserve, fomc, fed, percent, rate, ...
	T6	climate, climate change, green, risk, transition, climate relate, carbon, ...
	T7	islamic, islamic finance, islamic financial, malaysia, shariah, finance, sukuk, ...
	T8	japan, price, percent, growth, rate, increase, firm, ...
	T9	bsp, financial inclusion, inclusion, philippine, banking, service, sector, ...
12	T0	monetary, monetary policy, rate, inflation, stability, price, price stability, ...
	T1	risk, capital, banking, crisis, basel, institution, supervision, ...
	T2	euro, euro area, area, european, ecb, union, country, ...
	T3	inclusion, financial inclusion, service, banking, sector, albania, education, ...
	T4	payment, digital, service, card, cash, fintech, settlement, ...
	T5	federal, federal reserve, reserve, fomc, percent, fed, labor, ...
	T6	inflation, price, rate, growth, cent, target, high, ...
	T7	islamic, islamic finance, islamic financial, malaysia, shariah, finance, sukuk, ...
	T8	japan, price, percent, rate, firm, growth, increase, ...
	T9	bsp, philippine, percent, bangko, bangko sentral, sentral, pilipina, ...
	T10	climate, climate change, green, risk, transition, climate relate, carbon, ...
	T11	country, global, cent, exchange, growth, exchange rate, foreign, ...
15	T0	monetary, monetary policy, rate, stability, price, price stability, ecb, ...
	T1	risk, banking, capital, crisis, basel, institution, supervision, ...
	T2	euro, euro area, area, european, ecb, union, country, ...
	T3	financial inclusion, inclusion, service, banking, sector, financial service, india, ...
	T4	payment, digital, service, card, cash, settlement, innovation, ...
	T5	federal, federal reserve, reserve, fomc, percent, fed, labor, ...
	T6	inflation, price, rate, target, monetary policy, inflation target, monetary, ...
	T7	islamic, islamic finance, islamic financial, malaysia, shariah, finance, islamic banking, ...
	T8	japan, price, percent, japan economy, rate, firm, increase, ...
	T9	bsp, philippine, percent, bangko, bangko sentral, sentral, pilipina, ...
	T10	climate, climate change, green, risk, transition, climate relate, carbon, ...
	T11	country, exchange, global, exchange rate, foreign, international, currency, ...
	T12	albania, albanian, albanian economy, banking, development, supervisory council, fullani, ...
	T13	hong kong, kong, hong, hkma, rmb, mainland, china, ...
	T14	cent, growth, rate, canada, household, labour, productivity, ...

Continued on next page

K	Topics and top terms	
20	T0	inflation, rate, price, target, inflation target, expectation, monetary policy, ...
	T1	risk, basel, capital, banking, regulatory, supervision, regulation, ...
	T2	european, union, europe, euro, country, crisis, monetary union, ...
	T3	growth, cent, productivity, labour, rate, increase, high, ...
	T4	payment, digital, service, card, innovation, cash, settlement, ...
	T5	federal, federal reserve, reserve, fomc, percent, fed, labor, ...
	T6	euro, euro area, area, ecb, govern council, growth, council, ...
	T7	islamic, islamic finance, islamic financial, malaysia, shariah, finance, islamic banking, ...
	T8	japan, price, percent, japan economy, firm, yen, rate, ...
	T9	bsp, philippine, percent, bangko, bangko sentral, sentral, pilipina, ...
	T10	climate, climate change, green, risk, transition, climate relate, carbon, ...
	T11	south, africa, south africa, african, cent, south african, rand, ...
	T12	financial inclusion, inclusion, service, banking, sector, india, financial service, ...
	T13	hong kong, kong, hong, hkma, rmb, mainland, china, ...
	T14	credit, loan, liquidity, asset, mortgage, bond, rate, ...
	T15	albania, albanian, albanian economy, banking, development, supervisory council, fullani, ...
	T16	monetary, monetary policy, rate, stability, price stability, price, ecb, ...
	T17	global, exchange, country, exchange rate, foreign, international, crisis, ...
	T18	canada, canadian, cent, canadian economy, global, canadian dollar, target, ...
	T19	swiss, snb, franc, switzerland, swiss franc, swiss national, rate, ...

B LDA Candidate Topics

Table 11 reports the highest-probability words obtained by LDA for each candidate number of topics. This appendix complements the LDA model selection discussion in the main text. It shows that small values of K merge several broad macro-financial themes, while larger values of K improve perplexity and log-likelihood but lead to more diffuse or fragmented topic structures. The final choice $K = 10$ is therefore an interpretability-driven compromise.

Table 11: Top LDA terms for all candidate values of K. Each topic is represented by its seven highest-probability terms.

K	Topics and top terms	
4	T0	inflation, rate, growth, price, monetary, monetary policy, euro, ...
	T1	risk, banking, crisis, capital, institution, stability, european, ...
	T2	rate, monetary, monetary policy, reserve, price, federal, asset, ...
	T3	payment, service, sector, development, country, business, growth, ...
6	T0	growth, country, global, investment, sector, increase, high, ...
	T1	risk, banking, capital, institution, management, regulatory, crisis, ...
	T2	reserve, rate, asset, credit, liquidity, risk, federal, ...
	T3	payment, service, banking, business, development, sector, provide, ...
	T4	euro, area, monetary, euro area, european, ecb, country, ...
	T5	inflation, rate, price, monetary, monetary policy, growth, increase, ...
8	T0	growth, country, global, investment, world, sector, high, ...
	T1	risk, banking, capital, institution, crisis, stability, regulatory, ...
	T2	exchange, rate, liquidity, reserve, currency, foreign, asset, ...
	T3	credit, loan, asset, debt, household, business, mortgage, ...
	T4	euro, area, euro area, european, monetary, ecb, country, ...
	T5	price, rate, growth, inflation, increase, continue, monetary, ...
	T6	inflation, rate, monetary, monetary policy, price, target, low, ...
	T7	payment, service, banking, development, sector, business, industry, ...
10	T0	growth, increase, high, productivity, labour, investment, low, ...
	T1	risk, banking, capital, institution, crisis, regulation, regulatory, ...
	T2	reserve, federal, liquidity, federal reserve, asset, rate, fund, ...
	T3	credit, loan, debt, asset, household, rate, mortgage, ...
	T4	euro, area, euro area, european, monetary, ecb, country, ...
	T5	price, rate, growth, inflation, increase, continue, percent, ...
	T6	inflation, rate, monetary, monetary policy, price, target, low, ...
	T7	payment, service, banking, digital, money, technology, customer, ...
	T8	exchange, country, global, foreign, international, rate, exchange rate, ...
	T9	sector, development, climate, work, institution, business, finance, ...

Continued on next page

K	Topics and top terms	
12	T0	growth, fiscal, country, public, government, increase, high, ...
	T1	risk, capital, banking, crisis, institution, stability, financial stability, ...
	T2	reserve, federal, federal reserve, liquidity, rate, asset, fund, ...
	T3	credit, loan, asset, banking, lending, risk, sector, ...
	T4	euro, area, euro area, european, ecb, monetary, country, ...
	T5	rate, monetary, price, monetary policy, inflation, exchange, exchange rate, ...
	T6	inflation, monetary, monetary policy, rate, price, target, expectation, ...
	T7	payment, service, banking, digital, money, technology, industry, ...
	T8	global, country, exchange, international, foreign, capital, growth, ...
	T9	work, climate, change, sector, people, development, business, ...
	T10	japan, percent, increase, price, investment, continue, development, ...
	T11	growth, price, increase, high, rate, rise, demand, ...
15	T0	growth, fiscal, government, public, debt, increase, gdp, ...
	T1	risk, capital, banking, crisis, stability, institution, financial stability, ...
	T2	liquidity, bond, risk, fund, asset, rate, financial market, ...
	T3	credit, loan, asset, risk, lending, mortgage, banking, ...
	T4	euro, area, euro area, european, ecb, monetary, country, ...
	T5	rate, inflation, monetary, monetary policy, price, exchange, exchange rate, ...
	T6	inflation, monetary, monetary policy, rate, price, target, term, ...
	T7	federal, reserve, federal reserve, board, percent, community, fed, ...
	T8	global, country, exchange, international, foreign, capital, domestic, ...
	T9	work, people, world, country, change, great, think, ...
	T10	japan, price, percent, rate, increase, firm, continue, ...
	T11	growth, price, rate, increase, high, inflation, rise, ...
	T12	risk, climate, pandemic, change, global, transition, green, ...
	T13	payment, service, digital, technology, money, customer, innovation, ...
	T14	sector, banking, development, service, finance, institution, industry, ...
20	T0	fiscal, government, debt, public, country, growth, crisis, ...
	T1	crisis, stability, risk, financial stability, banking, regulation, sector, ...
	T2	board, information, governance, institution, committee, work, reserve, ...
	T3	credit, loan, mortgage, lending, rate, banking, household, ...
	T4	euro, area, euro area, european, ecb, monetary, union, ...
	T5	exchange, rate, exchange rate, country, currency, foreign, international, ...
	T6	monetary, monetary policy, inflation, rate, price, target, expectation, ...
	T7	federal, federal reserve, reserve, percent, community, labor, fomc, ...
	T8	global, growth, country, emerge, domestic, trade, sector, ...
	T9	country, world, people, work, think, come, know, ...
	T10	investment, bond, fund, investor, corporate, capital, company, ...
	T11	growth, productivity, increase, high, labour, low, rate, ...
	T12	climate, risk, pandemic, change, climate change, green, transition, ...
	T13	payment, service, digital, technology, innovation, customer, datum, ...
	T14	sector, banking, development, service, finance, islamic, industry, ...
	T15	money, reserve, india, settlement, currency, payment, security, ...
	T16	risk, capital, banking, management, firm, basel, regulatory, ...
	T17	liquidity, asset, risk, reserve, rate, crisis, credit, ...
	T18	inflation, price, rate, growth, high, increase, monetary, ...
	T19	japan, price, rate, percent, increase, monetary, firm, ...

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